Dialect Preservation or Normalization? A Cross-Paradigm Diagnosis of German Dialect ASR Bias

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Abstract

There is growing evidence that large-scale language and speech models encode systematic bias towards standard written varieties, often at the expense of dialectal or non-standard forms. We investigate this phenomenon in Automatic Speech Recognition (ASR) for German dialects, focusing on how different model families "choose" between dialect preservation and enforced normalization. Using a new diagnostic metric—Dialect Preference Score (DPS)—and thorough behavioral analysis across two complementary datasets, we reveal how model architecture, tokenization, and priors interact to shape this bias. Our results expose elementary design decisions that drive standardization, challenging dominant evaluation conventions and motivating better behavioral metrics.

1 Introduction

Bias towards standard written language is a persistent challenge in both text and speech processing. Recent work shows that large language models, when faced with non-standard input, often default to standard language forms, reinforcing linguistic stereotypes and marginalizing dialectal varieties (Fleisig et al., 2024). On text generation tasks, Deas et al. (Deas et al., 2023) demonstrate that language models perform less reliably on African American English than on mainstream English, usually reverting to "safer" standard expressions. Such tendencies extend to other diglossic contexts. For example, ChatGPT and related models perform far worse on Arabic dialects than on Modern Standard Arabic (Khondaker et al., 2023), while transparent preference for standard entities is observed more broadly across languages and cultures (Naous et al., 2023). Despite being widely known in the NLP community, these issues remain underexplored in speech and especially ASR for dialects and non-standard varieties.

In this study, we ask: **How do state-of-the-art ASR models transcribe dialectal speech—do they preserve non-standard forms, or do they systematically rewrite them into the standard variety?** We focus on a rigorous behavioral diagnosis, shifting the evaluation paradigm from error counts to preference measurement, and proposing new metrics that reveal architectural bias.

2 Related Work

Bias diagnosis and measurement. Difference-score bias metrics have become standard in NLP. Nadeem et al. introduced StereoSet, which quantifies language model bias by contrasting stereotypical and anti-stereotypical completions (Nadeem et al., 2021). Unlike regular accuracy metrics, difference-scores expose directional preference. In speech, most ASR work still uses Word Error Rate (WER), which—as Cotterell et al. and others have shown—penalizes legitimate dialectal variants and is poorly suited to morphologically complex or non-standardized languages (Cotterell et al., 2018).

Dialectal ASR. The mismatch between dialect and standard forms has received growing attention in the speech community. Nigmatulina et al. examined models trained on dialectal versus normalized transcriptions for Swiss German, finding that standard WER underestimates dialectal faithfulness and drives systems toward normalization (Nigmatulina et al., 2020). Multilingual ASR benchmarks underline that model architecture—CTC (Connectionist Temporal Classification) versus encoder-decoder—shapes how strongly decoding is influenced by priors versus acoustic cues (Nahabwe et al., 2025).

LLM critique and behavioral auditing. Auditing model preferences has become a research frontier. For text, Wang et al. proposed Shepherd, a critic model for LLM output critique (Wang et al., 2023). For speech, holistic behavioral indices like

DPS help to reveal model tendencies not captured by accuracy-focused scoring.

3 Datasets and Methodology

Our analysis uses two complementary German dialect ASR datasets. SwissDial provides parallel dialect and standard references for eight Swiss German dialects, enabling direct measurement of preference between non-standard and standard forms (Dogan-Schönberger et al., 2021). Betthupferl contains diverse Southern German dialects—Franconian, Bavarian, Alemannic—with historically non-codified and variable "pronunciation spelling" transcriptions, increasing annotation noise and reflecting realistic out-of-standard conditions (Blaschke et al., 2025).

We compare three model types: (1) CTC-based models (e.g., wav2vec2 XLS-R), which rely primarily on acoustic cues; (2) encoder-decoder (e.g., Whisper), which introduce strong language model priors; (3) LLM-ASR, which integrate large-scale decoding head (e.g., Qwen3-ASR).

For dialectal/standard proximity, we introduce the Dialect Preference Score (DPS):

$$\text{Sim}_{\text{dial}} = 1 - \frac{\text{Edit}(P, T_{\text{dial}})}{\max(|P|, |T_{\text{dial}}|)}, \quad (1)$$

$$\text{Sim}_{\text{std}} = 1 - \frac{\text{Edit}(P, T_{\text{std}})}{\max(|P|, |T_{\text{std}}|)} \quad (2)$$

where P is the prediction, T_{dial}, T_{std} are reference transcriptions, and $\text{Edit}(\cdot, \cdot)$ is the character-level Levenshtein distance. The DPS is defined as:

$$\text{DPS} = \text{Sim}_{\text{dial}} - \text{Sim}_{\text{std}} \quad (3)$$

Positive DPS means dialectal alignment, negative means normalization. We ignore degenerate cases where both similarities fall below a threshold, $\max(\text{Sim}_{\text{dial}}, \text{Sim}_{\text{std}}) < 0.1$, following defense against survival bias.

We use signal-to-noise ratio (SNR) augmentation to systematically disrupt acoustic input, evaluating model robustness and prior reliance.

4 Results

Paradigm separation and scaling effects. Figure 1 displays comprehensive model-wise behavioral distributions (DPS on SwissDial). CTC-based systems consistently achieve positive DPS, reflecting preservation of dialectal evidence. In contrast,

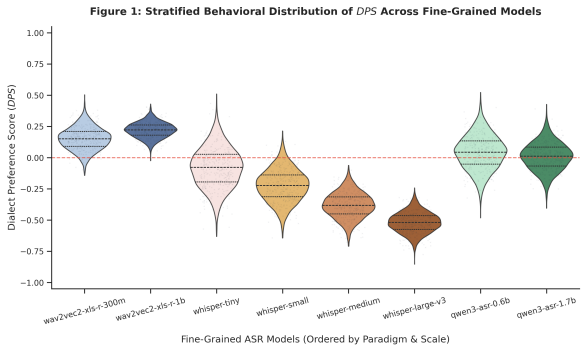


Figure 1: DPS distribution across fine-grained models on SwissDial. Clear regime separation: CTC favors dialect retention, Whisper drives normalization, LLM-ASR mixes both.

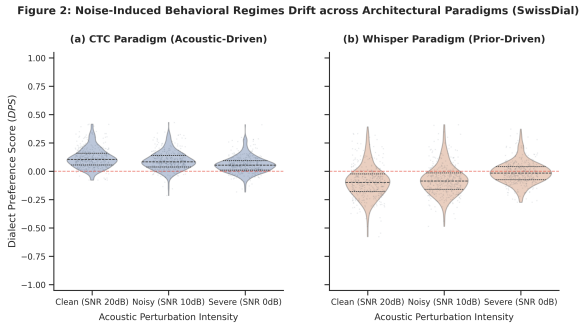


Figure 2: Noise-induced regime drift: Under lower SNR, encoder-decoder models amplify normalization bias toward standard forms.

encoder-decoder models show steady and systematic normalization, with scaling (e.g., Whisper Tiny → Large-v3) intensifying this effect as model priors gain dominance. LLM-ASR models typically yield near-zero DPS, often producing "hybrid" transcriptions that combine dialectal and standard segments.

Noise-induced regime drift. As SNR decreases (Figure 2), encoder-decoder models (Whisper) increasingly rely on text priors. When acoustic cues diminish, their outputs further drift toward standard German. In contrast, CTC-based systems preserve dialectal patterns, though variance increases with noise. These observations confirm and extend findings on model prior reliance in low-resource ASR (Nahabwe et al., 2025).

Cross-dataset findings and evaluation failures. On Betthupferl, all model types suffer a shift toward negative DPS, highlighting the confound introduced by historical annotation norms and inconsistent spelling. Notably, standard WER monotonically penalizes dialectal divergence, echoing claims that conventional scoring cannot capture nuanced

model behavior in non-standard settings (Cotterell et al., 2018; Nigmatulina et al., 2020).

5 Discussion

Our study reveals that the architecture and training regime of ASR systems drive their behavioral outcomes, even before considering training set size or diversity. The CTC pipeline, devoid of strong priors, is acoustically "honest" in its dialect transcription. Encoder-decoder and LLM-ASR families, by contrast, instantiate powerful priors through tokenization and decoding—an effect closely paralleling cultural and linguistic bias in text generation (Naous et al., 2023). When noise or annotation ambiguity grows, standardization bias accelerates, raising urgent questions for real-world multilingual, dialect-rich contexts.

This behavioral diagnosis challenges the widespread use of WER as an "objective" measure for dialectal ASR and motivates new indices, such as DPS, that are robust to annotation noise and reveal systematic preference. Our findings extend and reinforce previous claims from dialectal NLP, now grounded in speech and acoustic inference.

6 Conclusion

This work presents a cross-paradigm behavioral analysis of dialectal bias and normalization in German ASR. Our analysis demonstrates, using metrics like DPS and a controlled experimental suite, that model family and decoding architecture—not just training data—fundamentally shape the fate of dialectal input. We urge the community to adopt behavioral metrics and stress-test evaluation protocols for linguistic accessibility and fairness in speech technology.

Acknowledgments

Anonymized for submission.

A Additional Results

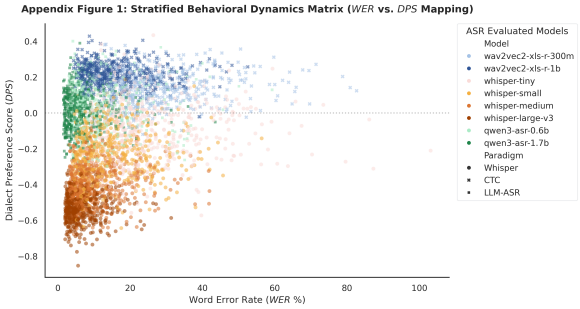


Figure 3: Word Error Rate (WER) vs. DPS: No fixed link between error rate and behavioral bias; demonstrates metric independence.

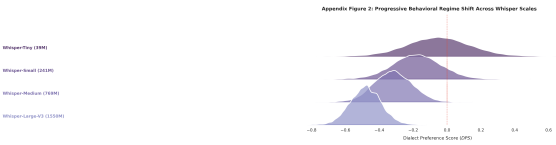


Figure 4: Whisper scaling (Tiny to Large-v3) and DPS: Larger models amplify normalization bias.

References

Verena Blaschke, Miriam Winkler, Constantin Förster, Gabriele Wenger-Glemser, and Barbara Plank. 2025. Betthupferl: A multi-dialectal dataset for German dialect ASR and dialect-to-standard speech translation. In *Interspeech*.

Ryan Cotterell, Sabrina J. Mielke, Jason Eisner, and Brian Roark. 2018. Are all languages equally hard to language-model? In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*.

Nicholas Deas, Jessi Grieser, Shana Kleiner, Desmond Patton, Elsbeth Turcan, and Kathleen McKeown. 2023. Evaluation of African American language bias in natural language generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Pelin Dogan-Schönberger, Julian Mäder, and Thomas Hofmann. 2021. SwissDial: Parallel multidialectal corpus of spoken Swiss German. In *Interspeech*.

Eve Fleisig, Genevieve Smith, Madeline Bossi, Ishita Rustagi, Xavier Yin, and Dan Klein. 2024. Linguistic bias in ChatGPT: Language models reinforce dialect discrimination.

Md Tawkat Islam Khondaker, Abdul Waheed, El Moatez Billah Nagoudi, and Muhammad Abdul-Mageed. 2023. GPTAraEval: A comprehensive evaluation of ChatGPT on Arabic NLP. *arXiv preprint*.

Moin Nadeem, Anna Bethke, and Siva Reddy. 2021. StereoSet: Measuring stereotypical bias in pretrained language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Alvin Nahabwe, Sulaiman Kagumire, Denis Musinguzi, Bruno Beijuka, Jonah Mubuuke Kyagaba, Peter Nabende, Andrew Katumba, and Joyce Nakatumba-Nabende. 2025. Benchmarking automatic speech recognition models for African languages. In *Proceedings of Machine Learning Research (Deep Learning Indaba)*.

Tarek Naous, Michael J. Ryan, Alan Ritter, and Wei Xu. 2023. Having beer after prayer? measuring cultural bias in large language models. *arXiv preprint*.

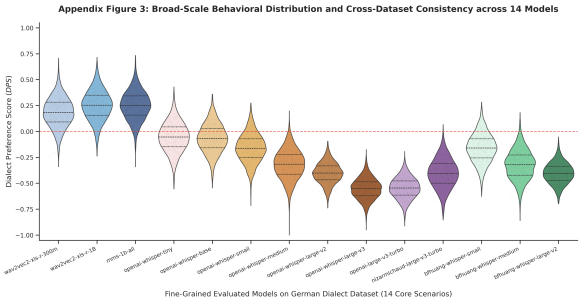


Figure 5: Cross-dataset comparison on Betthupferl: CTC shows positive DPS, Whisper and LLM-ASR retain negative drift.

Iuliia Nigmatulina, Tannon Kew, and Tanja Samardžić. 2020. ASR for non-standardised languages with dialectal variation: the case of Swiss German. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties and Dialects*.

Tianlu Wang, Ping Yu, Xiaoqing Ellen Tan, Sean O'Brien, Ramakanth Pasunuru, Jane Dwivedi-Yu, Olga Golovneva, Luke Zettlemoyer, Maryam Fazel-Zarandi, and Asli Celikyilmaz. 2023. Shepherd: A critic for language model generation. *arXiv preprint*.